# Calcurion Maths Worksheets

A-Level Mathematics

**Topic:** Pure 1 – Exponentials & Logarithms

Medium & Hard • 25 Questions

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## Section A – Medium (Q1–15)

**1.** Solve $3^x = 9$. **[2]**

**2.** Solve $2^x = 32$. **[2]**

**3.** Solve $5^x = 1$. **[2]**

**4.** Solve $4^x = 8$. **[3]**

**5.** Solve $9^x = 27$. **[3]**

**6.** Evaluate $\log_2 8$. **[2]**

**7.** Evaluate $\log_3 81$. **[2]**

**8.** Evaluate $\log_5 1$. **[2]**

**9.** Solve $\log_2 x = 5$. **[2]**

**10.** Solve $\log_3 x = 4$. **[2]**

**11.** Solve $\log_x 16 = 2$. **[3]**

**12.** Express $\log 12 + \log 3$ as a single logarithm. **[2]**

**13.** Express $\log 20 - \log 4$ as a single logarithm. **[2]**

**14.** Solve $2^{x+1} = 16$. **[3]**

**15.** Solve $3^{2x} = 27$. **[3]**

## Section B – Hard (Q16–25)

**16.** Solve $2^x + 2^{x+1} = 12$. **[4]**

**17.** Solve $3^{2x} - 4 \cdot 3^x + 3 = 0$. **[5]**

**18.** Solve $\log_2(x+1) + \log_2(x-1) = 3$. **[5]**

**19.** Solve $\log_3(x+2) + \log_3 x = 1$. **[5]**

**20.** Solve $e^{2x} - 3e^x + 2 = 0$. **[5]**

**21.** Express $2\log x + \log 3 - \log 5$ as a single logarithm. **[3]**

**22.** Solve $\log_2 x + \log_2(x-2) = 3$. **[5]**

**23.** The population $P = 100 \times 2^{t/10}$. Find $P$ when $t = 30$ and the time when $P = 800$. **[5]**

**24.** Solve $5^x = 3^{x+1}$. Give answer in terms of ln. **[4]**

**25.** Solve $\ln(x+1) - \ln x = 1$. **[4]**

*End of Worksheet* • *Show all working clearly.*